MIL-T-49435(ER)
19 January 1983

MILITARY SPECIFICATION

TEST SET, LASER DETECTING
AN/UPM-148()

This specification is approved for use by the U.S. Army Electronics Research and Development Command (ERADCOM), Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE.

1.1 Scope. This specification covers the requirements for the construction and testing of the Test Set, Laser Detecting AN/UPM-148(). The AN/UPM-148() is a special test equipment for the Laser Detecting Set AN/AVR-2(). The AN/UPM-148(), hereinafter called Test Set (TS), consists of the following major components:

Simulator, Laser Detecting	SM-815()/UPM-148
Case	
Cable Assembly, Power	
Control and Display Assembly	

Beneficial comments, (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, ERADCOM, Fort Monmouth, N.J. 07703, ATTN: DELEW-ES-PE, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 6625

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2. APPLICABLE DOCUMENTS

2.1 Issues of documents. The effective issue or revision of the following documents shall be that listed in the Department of Defense Index of Specifications and Standards (DODISS) and supplements thereto, unless (i) specific issues are set forth therefore in the cited specifications or (ii) different issues than those specified in the cited specificatons are set forth in the solicitation. The date of the applicable DODISS and supplements thereto shall be as specified in the solicitation or contract.

SPECIFICATIONS:

MIL-P-116	Preservation-Packaging, Method of
MIL-P-11268	Parts Materials and Processes Used in Electronic Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-T-28800	Test Equipment for use with Electrical and Electronic Equipment, General Specifications for
MIL-E-55585	Electronics Equipment and Parts, Packaging of

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MIL-D-49432(ER) Detecting Set, Laser AN/AVR-2()

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD 454	Standard General Requirements for Electronic Equipment
MIL-STD-461 w/Notice 4(EL)	Electromagnetic Interference Characteristics, Requirements for Equipment
MIL-STD-462 w/Notice 3(EL)	Electromagnetic Interference Characteristics, Measurements of
MIL-STD-726	Packaging Requirement Codes
MIL-STD-781	Reliability Tests: Exponential Distribution
MIL-STD-810	Environmental Test Methods
MIL-STD-1472	Human Engineering Design, Criteria for Military Systems, Equipment and Facilities
MIL-STD-1562	Lists of Standard Microcircuits

DRAWINGS

ERADCOM

DLC5072700	Test Set, Laser Detecting AN/UPM-148
DLC5072710	Simulator, Laser Detecting SM-815/UPM-148

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DRAWINGS (Cont'd)

C5072711	Acceptance Test Procedure for SM-815/UPM-148
C5072701	Acceptance Test Procedure for AN/UPM-148

(Copies of specifications, standards, drawings and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies).

2.2 Other Publications. The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

NATIONAL FIRE PROTECTION ASSOCIATION, INTERNATIONAL

National Fire Codes, Volume 6, Electrical

(Application for copies should be addressed to National Fire Protection Association, International, 470 Atlantic Avenue, Boston, Mass. 12210.)

AMERICAN NATIONAL STANDARDS INSTITUTE

ANSI Z35.1	Accident Prevention Signs, Specification for

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, N.Y. 10018.)

DEPARTMENT OF LABOR OCCUPATIONAL SAFETY AND HEALTH ADMINISTRATION

Title 29, Code of Federal Regulations, Chapter XVII, Part 1910, Occupational Safety and Health Standards

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington, D.C. 20402)

NATIONAL INSTITUTE FOR OCCUPATIONAL SAFETY AND HEALTH (NIOSH) Toxic Substance Annual List

(Application for copies should be addressed to the U.S. Department of Health, Education and Welfare, Rockville, Maryland, 20852.)

3. REQUIREMENTS.

3.1 Functional description. The TS provides the means for detecting and isolating faults in the AN/AVR-2. The TS is compatible with all operating parameters of the AN/AVR-2 as specified in MIL-D-49432(ER) and no degradation in the AN/AVR-2 performance occurs as a result of tests. The TS conforms to the requirements of MIL-T-28800.

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3.2 First article. When specified in the contract or purchase order, the contractor shall furnish first article units in accordance with 4.4 (See 6.6).

3.3 Construction. The equipment shall be constructed in accordance with the requirements of this specification and the drawings listed in USAERADCOM Data List, DLC5072700, which describes the TS including the requirements for parts, materials and processes, thereon. The following listed items are all part of the TS:

DLC5072710	Simulator, Laser Detecting
C5072730	Cable Assembly, Power
C5072705	Case
C5072790	Control and Display Assembly

3.3.1 Case. The case, style C enclosure, shall meet the requirements of 3.7.3, MIL-T-28800.

3.4 Parts, materials and processes; general. In addition to the requirements of this specification, the general requirements of MIL-P-11268(EL), including the selection requirements therein, shall apply.

3.4.1 Finish. The equipment shall be finished in accordance with MIL-F-14072(EL) and the equipment drawings.

3.5 Marking.

3.5.1 General. Marking shall conform to MIL-M-13231(EL) (see 4.5). Front panel marking shall be Group I as described in MIL-M-13231(EL). Location of marking shall be as specified on the equipment drawings.

3.5.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.3 Serial numbers. The following items shall have serial numbers:

C5072700	Test Set, Laser Detecting
C5072710	Simulator, Laser Detecting
C5072725	Circuit Card Assembly
C5072720	Circuit Card Assembly
C5072735	Power Supply
C5072705	Case

3.6 Performance.

3.6.1 Primary power. The TS shall maintain specified performance herein, with an input voltage range of 105 to 130 VAC at 50 to 60 Hz. The TS shall provide a minimum of 125% of the bench test power required by the AN/AVR-2. The power requirement for the AN/AVR-2 shall be 234 watts ±5%

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3.6.1.1. Primary power, Simulator. The TS shall provide 117 ± 5 VAC input voltage for the simulator SM-815/UPM-148.

3.6.2 Signal source. The TS shall be capable of selecting the signal source of the simulator for illuminating the AN/AVR-2 sensors.

3.6.2.1 Signal source control and monitoring. The TS shall be capable of energizing the selected signal source of the simulator and verifying its operation.

3.6.3 Pulse repetition rates (PRR). The TS shall generate the same PRR as specified in 3.1 of MIL-D-49432 (ER).

3.6.4 Circuit card assembly (CCA) Functional Check. The TS with standard test equipment shall be capable of functionally checking the CCA's of the AN/AVR-2 for proper operation.

3.6.5 Simulator performance.

3.6.5.1 Optical signal sources. The simulator shall provide optical signal sources in 3 Bands in accordance with C5072711 for Band I, C5072435 for Band II and C5072421 for Band III.

3.6.5.2 Beam pattern. The output of the sources for all bands shall be in accordance with drawings of 3.6.5.1.

3.6.5.3 Pulse repetition frequency (PRF) control. The simulator shall provide PRF test signals for Bands I & II. Band III shall provide a fixed rate PRF.

3.6.5.4 Self-Test. The simulator shall have an indication on the TS panel to indicate when test signals are being radiated.

3.7 Environmental requirements. The equipment shall meet the following service conditions without degradation in specified performance. Where a test is referenced, meeting the performance criteria shall be considered as compliance with the requirements.

3.7.1 Temperature.

3.7.1.1 Low Temperature. The TS shall be capable of operation at ambient temperatures as low as +32°F and of being stored at temperatures as low as -70°F. Before, during, and after testing as specified in 4.9.1.1 the equipment shall meet the performance specified in 3.6.

3.7.1.2 High temperature. The TS shall be capable of operation at ambient temperatures as high as +125°F and of being stored at temperatures as high as +160°F. Before, during, and after testing as specified in 4.9.1.2 the equipment shall meet the performance specified in 3.6.

3.7.2 Altitude. The TS shall be capable of operation at 10,000 feet altitude and of being air transported at 40,000 feet. Before, during, and after testing as specified in 4.9.2 the equipment shall meet the performance specified in 3.6.

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3.7.3 Humidity. The TS shall be operable without degradation in specified performance during exposure to extreme high humidity levels (95% RH at 86°F) and shall sustain no physical damage after prolonged exposure to extreme high humidity levels (to be determined by cycling from +70°F to +129°F at 94% RH) as encountered in tropical areas. Before, during, and after testing as specified in 4.9.3 the equipment shall meet the performance specified in 3.6. The equipment shall exhibit no physical damage such as corrosion, rust, blistering, swelling, or deterioration of parts and materials.

3.7.4 Vibration cycling. The TS shall be capable of operating and shall sustain no physical damage after being subjected to the vibrations normally encountered by equipment transported by rail, sea, truck, or semitrailer. Before and after testing as specified in 4.9.4 the equipment shall meet the performance specified in 3.6.

3.7.5 Bench handling. The TS shall be capable of operation and shall sustain no physical damage after exposure to the shock intensities normally encountered during servicing. Before and after testing as specified in 4.9.5 the equipment shall meet the performance specified in 3.6.

3.7.6 Fungus. No materials or components of the TS shall be nutrient to fungus as demonstrated by the test of 4.9.6. No materials shall be used other than inherently fungus resistant grades per Requirement 4 of MIL-STD-454.

3.7.7 Loose cargo bounce. The TS shall be capable of operation and shall sustain no major physical damage (cosmetic damage permitted) after being subjected to the shocks normally encountered during transportation as loose cargo. Before and after testing as specified in 4.9.7, the equipment shall meet the performance specified in 3.6.

3.7.8 Shock. The equipment in the case shall withstand shock induced by rough handling. Before and after testing as specified in 4.9.8 the equipment shall meet the performance specified in 3.6.

3.8 Electromagnetic interference. The TS shall comply with the emission and susceptibility requirements of MIL-STD-461, Notice 4(EL) listed below when tested as specified in 4.10:

CE02	CS02	RE02	RS02
CE03	CS06	RE02.1	RS03
CE04			RS03.1
CE05			

3.9 Reliability. The TS shall meet the lower test MTBF ($\theta 1$) of 350 hours (See 4.11).

3.10 Burn-in. Each TS to be delivered on contract, or to be subjected to first article tests in accordance with this specification shall be subjected to a 48 hour burn-in to limit infant mortality. The last 24 hours of the burn-in period shall be failure free. This burn-in shall be prior to any or all tests herein specified (See 4.3).

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3.11 Systems safety engineering.

3.11.1 Personnel hazards. Personnel hazards shall be kept to a minimum by control of configuration changes and parts selection to assure compliance with Requirement 1 of MIL-STD-454. Compliance with these requirements shall be verified through a visual inspection (see 4.14).

3.11.2 Edge rounding, exposed. Exposed edges shall be rounded to a minimum radius of 0.04 inch (1mm), and exposed corners to a minimum of 0.03 inch (0.75mm).

3.12 Human engineering. Configuration design criteria affecting human factors operation, maintainability and handling shall be maintained through compliance with MIL-STD-1472.

3.13 Interchangeability. Like units, assemblies, subassemblies and replaceable parts shall be physically and functionally interchangeable, without modification of such items or the system, and shall conform to Requirement 7 of MIL-STD-454 (see 4.13). Individual items shall not be selected for fit or performance, however, matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, or characteristic.

3.14 Workmanship. The equipment shall be manufactured and assembled in accordance with Requirement 9 of MIL-STD-454 and the applicable portions of the following paragraphs:

In MIL-P-11268(EL)

General requirements for plastic material and plastic parts
Wiring and cabling, including:
- Slack
- Protection
- Clearance
- Splicing

Connections, general
Grounding
- Wire and cable shielding

Flux and cleaning agent for soldering
Process for soldering
Cleaning
Riveting
General requirements for securing of parts
Cleaning after welding
Mechanical operation of controls
Process for welding
Self-tapping screws

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may utilize his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

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4.2 Classification of inspection. Inspection shall be classified as follows:

a. First article inspection (see 4.4 and 4.15).

b. Inspection covered by subsidiary documents (see 4.5).

c. Quality conformance inspection.

(1) Quality conformance inspection of equipment before packaging (see 4.6).

(2) Quality conformance inspection of packaging (see 4.14).

4.3 Burn-in. Prior to the start of any or all of the following tests, each TS shall be burned-in in accordance with 4.12 and shall meet the requirements of that procedure and paragraph 3.10 herein.

4.4 First article. Unless otherwise specified in the contract or purchase order, the first article inspection shall be performed by the contractor.

4.4.1 First article units. The contractor shall furnish five first article units of the complete TS.

4.4.2 First article inspection. The first article inspection shall consist of inspections specified in Table I and shall be performed in the order specified therein.

4.4.3 First article data. The first article test plan and test report shall be as required in the contract or purchase order (see 6.5).

4.5 Inspection covered by subsidiary documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification and the inspection requirements specified in the contract or purchase order.

ITEM	WHERE REQUIRED
Parts, materials, and processes; general	3.4
Finish	3.4.1
Marking	3.5
Safety	3.11
Human engineering	3.12

4.6 Quality conformance inspection of equipment before packaging. The contractor shall perform the inspections specified in 4.5 and 4.6.1 through 4.6.4 in that order. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements.

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TABLE I. First article inspection.

INSPECTIONS	REQ. PARA.	TEST PARA.	UNIT 1	UNIT 2	UNIT 3	UNIT 4-5
1. Inspection covered by subsidiary documents	3.4 3.5	4.5	Inspections to be performed on all units.			
2. Group A inspection	See Table II		Inspections to be performed on all units.			
3. Group B inspection	See Table III		Inspections to be performed on all units.			
4. Reliability	3.9	4.11				1
5. EMI	3.8	4.10			1	
6. Environmental						
Temperature	3.7.1	4.9.1	1			
Altitude	3.7.2	4.9.2	2			
Humidity	3.7.3	4.9.3	3			
Vibration	3.7.4	4.9.4		1		
Bench handling	3.7.5	4.9.5		2		
Fungus	3.7.6	4.9.6			2	
Loose cargo	3.7.7	4.9.7		3		
Shock	3.7.8	4.9.8		4		

NOTES: a. The inspections 1 through 3, in the order shown, shall be performed on all first article units before subjecting these units to any other inspection requirements specified in the table.

b. The numbers in the unit columns in the table covered by inspection specify the order of inspections for the indicated unit.

c. The equipment shall be thoroughly washed, cleaned, dried, and refurbished after the fungus test before proceeding with subsequent inspection, if required.

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4.6.1 Group A inspection. Each unit on contract or purchase order shall be inspected for conformance to the inspections specified in Table II. Discrete lots shall be formed from units that pass this inspection. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be subjected to sampling inspection, utilizing the procedures of MIL-STD-105, using general inspection level II, and AQL's indicated in Table II.

4.6.1.1 Order of inspection within Group A. Group A inspection shall be performed in an order satisfactory to the Government.

4.6.2 Group B inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjected to and met the Group A inspection. Interchangeability test samples may be separate units whose modules and replaceable parts were randomly selected and assembled prior to Group A testing. This inspection shall conform to Table III.

4.6.2.1 Sampling for Group B inspection. Sampling for Group B inspection shall be IAW MIL-STD-105D. Lot size shall be determined by the contractor.

4.6.2.2 Order of inspection within Group B. Group B inspection shall be performed in an order satisfactory to the Government.

4.6.3 Group C inspection. Group C inspection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspections specified in Table IV. Samples shall be selected in accordance with 4.6.3.1.

4.6.3.1 Sampling for Group C inspection. Three samples of equipment shall be selected at random from every 30 sets or fraction thereof produced. One different sample shall be used for each of the three Subgroup inspections in Table IV. Satisfactory Group C inspection of the first article samples shall satisfy the Group C inspection requirement for the first 30 units produced. Unless otherwise specified in the contract, the fungus test shall not be repeated after the first production lot Group C inspection unless parts, materials, or processes are changed.

4.6.3.2 Order of inspection within Group C. Group C inspection shall be performed in the order specified in Table IV.

4.6.3.3 Group C failures. Action required relative to Group C failures shall be as specified in the contract or purchase order.

4.6.4 Reinspection of conforming Group C sample units. Unless otherwise specified, sample units which have been subjected to and passed Group C inspection may be accepted on the contract provided all damage is repaired and the sample units are resubjected to and pass Group B and Group A inspection, in that order.

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TABLE II. Group A inspection

INSPECTION	REQ. PARA.	TEST PARA.	AQL MAJOR	AQL MINOR
Visual and Mechanical	3.3, 3.4, 3.5, 3.11 3.13, 3.14	4.5, 4.14	1%	1%
Primary Power	3.6.1	4.8	*1.5% for the Group	
Primary power, Simulator	3.6.1.1			
Signal Source	3.6.2			
Signal Source Control	3.6.2.1			
PRR	3.6.3			
CCA Functional Check	3.6.4	4.8		
Optical signal sources	3.6.5.1	4.8	*1.5% for the Group	
Beam Pattern	3.6.5.2			
PRF Control	3.6.5.3			
Self-Test	3.6.5.4	4.8		

* All electrical defects are considered major

TABLE III. Group B inspection

Inspection	REQ. PARA.	TEST PARA.	AQL
Interchangeability	3.13	4.13	4.0%

4.7 Group D inspection. This inspection shall consist of the tests specified in Table V and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection. Test conditions, methods, and procedures shall be in accordance with paragraph 4.11.

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TABLE IV. Group C inspection

Inspection	Req. Para.	Test Para.
Subgroup I		
Temperature	3.7.1	4.9.1
Altitude	3.7.2	4.9.2
Humidity	3.7.3	4.9.3
Subgroup II		
Vibration	3.7.4	4.9.4
Bench handling	3.7.5	4.9.5
Loose cargo	3.7.7	4.9.7
Shock	3.7.8	4.9.8
Subgroup III		
Fungus	3.7.6	4.9.6
EMI	3.8	4.10

TABLE V. Group D inspection

Inspection	Req. Para.	Test Para.
Reliability	3.9	4.11

4.7.1 Sampling for Group D inspection. 2 units shall be selected at random from the first 30 units. For subsequent Group D inspection, 2 units shall be selected at random from every 200 units or fraction thereof. Upon a rejection of a Group D Test and after the retest is subsequently passed, the sampling rate shall become 2 units selected at random from every 50 units until two consecutive reliability tests have been passed at which time the sampling rate shall revert back to 2 units from every 200 units or fraction thereof.

4.7.2 Failure definition. A failure occurrence during Group D inspection is defined as any deviation in performance from the requirements specified in paragraph 3.9.

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4.7.3 Reinspection of conforming Group D sample units. Unless otherwise specified, sample units which have been subjected to and passed Group D inspection may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.8 Performance. The TS shall meet all requirements of 3.6 when tested in accordance with the test procedures C5072701 and C5072711.

4.9 Environmental tests. All environmental test methods, conditions, and procedures given below are contained in MIL-STD-810. The equipment performance measurements shall be made under standard conditions as defined in MIL-STD-810. Successful completion of performance test specified herein shall be considered evidence of meeting the corresponding service condition.

4.9.1 Temperature. The TS shall be tested in accordance with the following test methods:

4.9.1.1 Low temperature. The TS shall be tested in accordance with Method 502.1, Procedure I of MIL-STD-810 and shall meet the requirements of 3.7.1.1.

Storage temperature (Step 2)---------- -70°F
Operating temperature (Steps 4 & 5)---------- +32°F
Make measurements at Steps 1,5 & 7

4.9.1.2 High temperature. The TS shall be tested in accordance with Method 501.1, Procedure I of MIL-STD-810 and shall meet the requirements of 3.7.1.2.

Storage temperature (Step 2, 3, 5)---------- +120°F
Storage temperature (Step 4)---------- +160°F
Operating temperature (Step 7 & 8)---------- +125°F
Make measurements at Steps 1,8 & 10

4.9.2 Altitude. The TS shall be tested in accordance with Method 500.1, Procedure I of MIL-STD-810, except Step 2 shall be 1500 feet. The rate of change of altitude shall not exceed 2000 feet per minute. A non-operating test shall be performed for one hour at 40,000 feet. The TS shall meet the requirements of 3.7.2 when tested at Steps 1, 2, and 4.

Operating altitude.....10,000 feet
Non-operating altitude.....40,000 feet

4.9.3 Humidity. The TS shall be tested in accordance with Method 507.1, Procedure II of MIL-STD-810, with the exception that Step 6 shall be limited to two 48 hour cycles. The TS shall be operated during the last 5 hours of the last cycle of Step 6, and shall meet the requirements of 3.7.3 when tested in Steps 4, 6 (last cycle only) and 8.

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4.9.4 Vibration cycling. The equipment, in its case, shall be tested in accordance with Method 514.2, Procedure X of MIL-STD-810, and shall meet the requirements of 3.7.4.

Equipment category.....g
Applicable table.....514.2-VII
Applicable figure.....514.2-7
Test curve.....AW

Make measurements before and after the test only. The frequency of applied vibration shall be swept logarithmically over the specified range.

4.9.5 Bench handling. The TS shall be tested in accordance with Method 516.2, Procedure V of MIL-STD-810, and shall meet the requirements of 3.7.5.

4.9.6 Fungus. Selected component parts of the equipment and samples of all materials, finishes, and processes used in the manufacture of the equipment shall be tested according to MIL-STD-810, Method 508.1, Procedure 1. No electronic operation shall be performed. At the conclusion of the test, component parts and samples shall show no fungal growth which is judged to interfere in any way with the intended operation of the equipment as a whole and shall meet the requirements of 3.7.6.

4.9.7 Loose cargo bounce. The TS shall be tested in accordance with Method 514.2, Procedure XI, Part 2 of MIL-STD-810, and shall meet the requirements of 3.7.7.

4.9.8 Shock. The equipment inside its case as prepared for field transport shall be tested according to Method 516.2, Procedure II of MIL-STD-810 except drop height shall be 12 inches (300 mm). The above procedure shall include 26 drops including 6 face drops, 8 corner drops and 12 edge drops.

4.10 Electromagnetic interference. The TS shall be tested in accordance with MIL-STD-462, Notice 3(EL). Upon completion of test, the TS shall meet the requirements of 3.8.

4.11 Reliability. The TS shall be tested in accordance with Test Plan XXIC of MIL-STD-781. The environmental conditions during reliability testing shall be the same as for burn-in (see paragraph 4.12.1).

4.11.1 Corrective action. Should the reliability test reach a reject decision, discontinue the test, immediately notify the procuring activity and promptly develop and propose a plan for correction of the deficiencies to the procuring activity. The specified performance and the required characteristics of the equipment will not be changed to achieve reliability requirements. Any corrective action will be proposed to the procuring activity as an engineering change proposal.

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4.12 Burn-in. Each TS submitted for acceptance shall be subjected to a 24-hour burn-in, as specified below. Each equipment which fails shall be repaired and returned to burn-in until it successfully survives 12 hours without a failure.

4.12.1 Burn-in conditions.

Temperature:	25°C
Vibration:	None
Voltage:	28Vdc ±5%
Duty Cycle:	100%

4.12.2 Burn-in procedure.

a. Mount TS on a bench in an area where temperature is controlled to 25°C ±5°C.

b. Set timer so that voltage cycles in accordance with the requirements of paragraph 4.11.

c. Continue above procedure for second 24 hour period.

d. At end of the burn-in perform Group A electrical test.

e. In the event of a test failure take necessary corrective action and resubject unit to one 12 hour burn-in cycle.

4.13 Interchangeability. The equipment shall be inspected to insure the physical and functional interchangeability requirements of 3.13.

4.13.1 Electrical interchangeability. All like modules and replaceable parts shall be tested for functional interchangeability by assembly of randomly selected parts. The assembly shall be tested to and shall meet Group A acceptance requirements. Electrical retuning or realignment of modules during production assembly may be expected and are permissible. Random selection for this assembly shall be supervised by the Government.

4.13.2 Mechanical interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.13. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of case, cover, and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

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c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as case, covers and mountings.

d. Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.14 Visual and mechanical inspection. The equipment shall be visually and mechanically inspected to insure conformance to the construction requirements of 3.3 and the workmanship requirements of 3.14. Defects shall be classified in accordance with MIL-STD-252(EL).

4.15 Packaging inspection. Packaging inspection requirements specified herein are classified as follows:

a. First Article Inspection of Packaging

b. Quality Conformance Inspection of Packaging

4.15.1 First Article Inspection of Packaging. Unless otherwise specified in the contract, First Article Inspection of Packaging shall be in accordance with the Unit Pack Design Validation Requirements of MIL-P-116.

4.15.2 Quality Conformance Inspection of Packaging.

4.15.2.1 Materials inspection. All materials to be used in packaging shall be inspected in accordance with the applicable material specification.

4.15.2.2 Preservation inspection. Inspection of preservation and interior markings shall be in accordance with group A and B Quality Conformance Inspection Requirements of MIL-P-116. Lot formation and sampling procedures shall be as specified therein.

4.15.2.3 Packing inspection. Inspection of packing and the marking for shipment and storage shall consist of the examinations specified in Table VI "PACKING INSPECTION PROVISIONS." Lot formation shall consist of all packs made of the same materials during an identifiable period and submitted at one time for acceptance. Sampling procedures shall be in accordance with MIL-STD-105, using a single sampling plan and Acceptable Quality Level of 4.0 percent defective.

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TABLE VI.

PACKING INSPECTION PROVISIONS

NO.	CHARATERISTIC	METHOD OF INSPECTION
101	Intermediate container not as specified	Visual
102	Improper closure of intermediate container	Visual
103	Shipping containers not in accordance with specification	Visual
104	Excessive cube	Visual
105	Improper blocking and bracing	Visual
106	Closure not in accordance with specification	Visual
107	Weight and size exceed container limitations	Weight & Measure
108	Strapping not in accordance with specification, incorrectly applied, omitted	Visual
109	Marking omitted, incorrect, or illegible	Visual

5. PACKAGING

5.1 Preservation. Preservation shall be as specified in MIL-STD-726, coded as follows:

5.1.1 Level A. JT-1-1-00-00-NR-X-ED-0-00-A

5.1.2 Level B. JS-1-1-00-00-NR-X-ED-0-00-B

5.2 Packing and marking. Packing and marking shall be in accordance with MIL-E-55585.

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6.0 NOTES

6.1 Intended Use. The Test Set, Laser Detecting, AN/UPM-148 is intended to be used at the AVIM maintenance level to fault isolate AN/AVR-2 components.

6.2 Ordering Data.

Procurement documents should specify:

a. Title, number and date of this specification and any amendment thereto.
b. Level A or Level B preservation and packing (see Section 5).

c. When first article inspection rough handling test are not required.

d. Marking and shipping requirements.

e. First article inspection.

(1) Number of units.
(2) First article test plan and test report(s) required.
(3) When first article packaging inspection test reports require Aquisition Activity approval prior to production unit packing.

f. Place of final inspection.

6.3 Definitions.

6.3.1 Temperature stabilization. Temperature stabilization will have been attained when the temperature of the part of the test item considered to have the longest thermal lag does not change more than 2.0°C (3.6°F) per hour.

6.3.2 Physical damage. Physical damage is defined as corrosion,rust, blistering, swelling, or other deterioration of parts and materials.

6.3.3 Fungus.

6.3.3.1 Growth colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.3.3.2 Microbial growth. Microbial growth is the growth of very minute organisms. Such organisms when present in large numbers may provide a colony visible to the naked eye.

6.3.3.3 Tubular germination. Tubular germination is partial growth by the production of hyphae, which are tubular shaped fungal filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

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6.3.4 Exfoliation. Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating or both of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

6.4 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.5 Data requirements. Deliverable data required by this specification is cited in 4.4.3 entitled First Article Data.

6.6 First article. First articles are required for inspection and approval (see 3.2 and 4.4).

6.7 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.8 Weight. For shipping purposes the weight of the TS shall not exceed 38 pounds.

6.9 Size. The size of the TS shall not exceed 20.6 inches wide by 10.0 inches deep by 13.6 inches high.

6.10 Level B preservation. When level B preservation is specified, this level of protection will only be used under known favorable conditions during transportation, storage and handling.

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